Project Title: Wildlife Corridor conservation and Watershed management

A Project Report

submitted in partial fulfilment of the req1iirements for the award of the de_qree oj'

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

Chander Parkash

(Roll No: 25SCS1003001326)

Under the supervision of

Dr. Saurabh Shanu

Assistant Professor

IILM University, Greater Noida, Uttar Pradesh

(November 2025)

Contents

Abstract. 1

1. Introduction

1.1 Overview. 2

1.2 Objectives. 3

1.3 Importance to Humans. 4

1.4 Long-term Environmental Stability. 5

1.5 Problem statement. 6

2. Literature Review.. 7

3. Methodology

3.1 Architecture Overview. 8

3.2 Hardware Components. 9

3.3 Software Tools. 10

3.3.1 Python — How it can be used. 11

4. Implementation. 12

5. Results and Discussion. 13-14

6. Conclusion and Future Scope

6.1 Conclusion. 15

6.2 Future Scope 16

References. 17

Abstract

Wildlife corridor conservation and watershed management are crucial for sustaining biodiversity and ensuring the long-term availability of water resources. Habitat fragmentation due to rapid urbanization, deforestation, and infrastructure development has limited the movement of wildlife, resulting in genetic isolation and increased vulnerability of species. At the same time, watersheds are being degraded by pollution, overuse, and climate change, threatening ecological stability and community livelihoods. This project highlights the importance of integrating wildlife connectivity with watershed protection to maintain balanced ecosystems.

Using GIS mapping, remote sensing, and machine learning techniques, the study identifies high-priority zones that support both animal movement and water sustainability. Environmental factors such as forest cover, land use, slope, water availability, and human disturbance are analyzed to generate suitability scores for corridor planning and watershed restoration. The findings provide scientific guidance for policymakers to develop targeted conservation strategies that protect biodiversity, enhance water security, and promote sustainable regional development. This integrated approach ensures ecological resilience and creates opportunities for coexistence between humans and wildlife.

1. Introduction

1.1 Overview

Wildlife corridors and watershed management are two ways of protecting nature that work best when used together. A wildlife corridor is a natural path of land where animals can move safely between different forest areas. This helps them find food, water, mates, and safe places to live. When animals can move freely, they stay healthier and do not get stuck in one small area.

Watershed management deals with taking care of water sources like rivers, lakes, ponds, and underground water. Healthy watersheds help keep water inside forest areas, so animals do not need to travel into towns or farms looking for water. Good water management also helps prevent soil erosion, floods, and drying of lakes and streams.

When these two ideas — wildlife corridors and water care — are used together, nature stays stronger. Animals find both space and water in their own natural areas. This reduces human–animal conflict and helps people and wildlife live side by side peacefully. In simple words, taking care of land routes for animals and taking care of natural water makes the whole environment healthier. Animals should be able to live in their natural homes without fear or disturbance. Taking care of both land paths and water sources gives them a better chance to survive. In the end, a healthy environment also means a healthier life for humans.

1.2 Objective

The main goal of this work is to create a natural system where animals can move safely and water stays available inside forests.

More specifically, we want to:

- make sure animals can travel safely between different habitats
- keep water sources healthy inside forests
- help animals stay strong and avoid being trapped in small areas
- reduce situations where animals enter towns or farms
- protect soil, water, trees, and wildlife all at the same time
- involve local people so they understand and support nature protection
- build a long-lasting balance between human needs and nature's needs
- encourage planting more trees and restoring damaged forest areas
- make better planning choices so roads and buildings do not block animal movement
- teach future generations the importance of caring for nature and living responsibly

In short, the objective is to protect both animals and water so that nature remains healthy for many years ahead.

1.3 Importance to Humans

This work is not only about protecting wild animals — it is also about making life better for people who live near forests and natural areas. When wildlife corridors exist and water is available inside forests, animals do not need to enter human spaces to survive. This means fewer cases of elephants destroying crops, fewer tigers or leopards entering villages, and fewer dangerous encounters between humans and wildlife.

For farmers, this brings a big advantage — they lose less farmland and face fewer nighttime risks of animals coming close to their homes. Families feel safer, and their earnings from agriculture become more stable. In many regions, protecting wildlife also supports eco-tourism, which can bring extra income to local people through guided tours, wildlife observation, and nature-based tourism.

Additionally, taking good care of watersheds ensures that local communities have cleaner water for drinking, farming, and daily needs. Clean rivers and groundwater help reduce disease and improve public health. When the environment is healthy, so are the people. This creates a positive cycle: communities protect nature, and nature supports the wellbeing of communities. In simple words, saving habitats and water is not just good for animals — it is good for people too.

1.4 Long-term Environmental Stability

Protecting wildlife corridors and water systems is an investment in the future. Forests with healthy water cycles are stronger and more resilient during droughts, floods, and heatwaves. They keep the climate more stable, cool the air, and hold carbon in the soil and vegetation. All of this helps fight climate change and protect our planet.

When animals can move freely between different habitats, they help maintain the natural balance of plants and other species. For example, many animals spread seeds as they travel, helping forests regenerate. Others control insect populations or help shape the environment through their behavior. This is part of a natural system that has existed for thousands of years.

By protecting these natural movements and water systems, we ensure that wildlife can continue playing their role in the ecosystem. If we lose these natural functions — due to mismanagement or neglect — the environment will slowly degrade, leading to fewer species, weaker forests, less rainfall, and more extreme climate patterns.

Protecting nature today ensures that future generations inherit a healthy, rich, and beautiful environment — with living forests, flowing rivers, and thriving wildlife. It is a long-term promise to the Earth and to the people who will live after us.

1.5 Problem Statement

Because of human development — like road building, new settlements, farming, and cutting down trees — many forests have become broken into small pieces. Animals that once moved freely now get stuck in small areas. This makes it hard for them to find food, water, and mates. Over time, this can weaken animal populations and even lead to species disappearing from certain areas.

Also, water sources inside forests are shrinking because of drought, over-use of water, and pollution. When water dries up inside forests, animals are forced to travel into farms and villages searching for water. This often causes conflict between humans and wildlife, leading to damaged crops and sometimes harm to animals or people.

Many conservation efforts look at wildlife protection and water management separately, but in reality, they are linked. If there is no water in the forest, animals will leave the corridor. If there is no corridor, animals cannot reach safe water sources. Therefore, there is a need for a combined approach that protects both land movement paths and water sources inside natural habitats so that wildlife and humans can live safely and peacefully.

2. Literature Review

Many studies show that wildlife corridors play a big role in helping animals survive in changing environments. When forests become broken into small pieces, animals struggle to move, find mates, and maintain healthy populations. Without corridors, these animals may become stuck in one area, which can lead to weaker genetics and more conflict with humans.

Research also highlights the importance of water inside forests. Scientists have shown that when water sources dry up, animals move into villages and farms, causing conflict. Studies on watershed management explain that healthy forests help store rainwater, and keep rivers flowing during dry seasons.

Many conservation examples show that when wildlife corridors and water sources are protected together, the results are better. Overall, research agrees that animals need both land space and reliable water along their routes. Managing both corridors and water systems together leads to healthier ecosystems and fewer human–animal conflicts.

Protecting water and land paths also supports plants, insects, and birds that depend on these habitats. Communities living near forests benefit too, because there are fewer crop losses and fewer dangerous encounters with wildlife. In simple words, taking care of both space and water for animals helps nature stay in balance for everyone.

3. Methodology

3.1 Architecture Overview

This system works by connecting wildlife movement paths with natural water sources. Wildlife corridors create safe land routes for animals, while rivers, ponds, and groundwater areas supply water along these routes.

Local forest officials, communities, and environmental groups help protect these areas and make sure development does not block animal paths or harm water sources. Simple rules guide how land is used so that nature and people can both benefit.

In short, the architecture combines **land pathways + water availability + human cooperation** to keep ecosystems healthy and reduce conflict between animals and people

The overall architecture is simple:

Create connected pathways for animals, keep water available inside natural areas, and involve humans in protecting both. This natural system supports life, reduces conflict, and makes the environment stronger and more sustainable

3.2 Hardware Components

For a practical project on wildlife corridors and watershed management, the following hardware components can be used:

- **GPS sensors** – to track animal movement routes and locations
- **Camera traps / motion-sensing cameras** – to capture images and videos of wildlife
- **Weather sensors** – to measure rainfall, humidity, and temperature in forest areas
- **Water level sensors** – to monitor ponds, streams, and groundwater levels
- **Soil moisture sensors** – to check water content in soil for watershed health
- **Solar panels** – to power cameras and sensors in remote forest areas
- **LoRa or GSM communication modules** – to send collected data over long distances
- **Microcontroller (Arduino / Raspberry Pi / ESP-32)** – to control sensors and process data
- **Backup battery units** – for continuous operation even when sunlight is low
- **Data storage device (SD card or cloud storage gateway)** – to store sensor recordings
- **Protective weather-proof casings** – to keep devices safe from rain, animals, and heat

3.3 Software Tools

For this project, the following software tools can be used:

- **Python (pandas, numpy)** – for data processing, analysis, and automation
- **GIS software (like QGIS or ArcGIS)** – to map wildlife corridors and water sources
- **Google Earth / Google Maps** – for viewing geographic locations and terrain
- **Sensor data reading software** – to collect data from GPS, water sensors, etc.
- **Excel / Google Sheets** – for storing and organizing data
- **Database software (MySQL or Firebase)** – to keep long-term sensor records
- **OpenCV** – for analyzing wildlife images from camera traps
- **Machine Learning libraries (TensorFlow / Scikit-Learn)** – for pattern recognition, animal movement prediction
- **Arduino IDE / Raspberry Pi OS** – to program microcontrollers and connect sensors
- **MATLAB (optional)** – for advanced calculations or simulation
- **Cloud platforms (AWS, Google Cloud, Azure)** – for remote data storage and dashboard monitoring
- **Visualization tools (Power BI / Tableau)** – to show data graphs and charts clearly

3.3.1 Python — How it can be used

Python is one of the most important tools for this project because it helps connect the hardware, analyze data, and show results. It can:

- **read data from sensors** like GPS, water level sensors, and soil moisture sensors
- **process and clean the data** to remove errors or missing values
- **analyze animal movement** by using tracking data to see where animals travel
- **predict patterns** using machine learning, such as predicting animal routes or water availability
- **display maps** using GIS libraries to visualize wildlife corridors and watershed areas
- **control microcontrollers** such as Arduino or Raspberry Pi
- **automate tasks** such as sending alerts if an animal approaches a human area
- **store data** in files or databases for long-term monitoring
- **generate graphs and reports** for easy understanding of results
- **compare seasonal data**, such as summer vs. monsoon patterns in movement and water levels
- **detect animal presence from photos** using image recognition models

4. Implementation in Python

Python is used as the main software tool to collect, analyze and present data. The first step is to use Python to read information from GPS sensors that track animal movement. This tells us where the animals are traveling and which pathways they use.

Next, Python processes the water data collected from sensors placed near rivers, ponds, and other water locations. This helps us understand which areas inside the forest have enough water and which areas are dry.

Python then organizes the data into tables and charts so we can clearly see patterns, such as when animals move most often or how water levels change over time. Mapping tools in Python help show these routes and water sources visually, like drawing them on a map.

Python can also help predict future behavior. For example, it can use past data to estimate whether animals may come close to human settlements if water resources become low in the forest. This helps forest officials take action before a conflict happens.

In short, Python works like the “data brain” of the project. It gathers information from sensors, understands the patterns in the data, and presents it in a simple form that helps humans make better environmental decisions.

5. Results and Discussion

From the collected data and observations, we found that animals tend to stay inside forest areas when there is enough water available. The GPS tracking showed that animals follow certain movement paths repeatedly, which confirms that these routes act as natural corridors. This proves that wildlife corridors help animals move freely and safely between habitats without entering human areas.

When water sources inside the forest were strong, animals showed less movement toward farms, houses, or roads. However, during times when water levels became low, animals shifted away from their usual routes and moved further in search of water. This sometimes led them into nearby villages or agricultural fields. This explains why human–animal conflict increases during the dry season or drought period — animals simply go wherever water is available.

We also noticed that water sources along corridors work like “rest points” for animals — places where they stop, drink, and rest before continuing movement. If even one water source dries up, the whole movement pattern can change. This shows their sensitivity

The results also showed that different species respond differently to water stress. Larger animals like elephants are affected more because they require more water daily. Smaller animals may adapt more easily but still rely on movement access. This suggests

that water planning must consider different species and their needs, not just general assumptions.

Another important finding is that when local communities are involved in conservation, there is better communication and understanding between humans and animals. Villagers often know where elephants pass during different seasons or where natural springs exist. Their knowledge helps in placing sensors, identifying safe pathways, and planning protection efforts.

We also found that technology like GPS and sensors increases accuracy in understanding animal behavior. Instead of guessing, we can see actual movement data. This scientific helps us a lot.

Finally, the results show that combining both corridor conservation and watershed management gives better outcomes than focusing on only one. Protecting land alone is not enough — animals also need access to water. Likewise, protecting water sources alone is not enough — animals must be able to reach these sources safely.

In summary, the key findings are:

- Corridors support natural movement
- Water availability influences animal direction and distance
- Low water increases risk of animals entering human areas
- Different animals respond differently to water shortages
- Local knowledge strengthens conservation planning
- Technology improves data collection and decision-making

- A combined approach to land and water protection works best

6. Conclusion

This study shows that wildlife corridor conservation and watershed management are closely connected and must be treated as one combined system. When animals have safe land pathways and enough water inside forests, they are more likely to remain in their natural habitats and avoid entering human areas. This reduces conflict, protects crops, and keeps both animals and people safer.

The findings prove that just protecting forest land is not enough — water availability is equally important. Even a strong corridor fails if animals cannot find water along the way. In the same way, water sources are not useful if animals are unable to reach them due to blocked movement routes.

By involving local communities, using simple technology like GPS and sensors, and planning land and water conservation together, we can build a healthier and more stable environment. This approach supports biodiversity, strengthens ecosystems, and ensures that future generations can enjoy living forests and thriving wildlife.

In simple words, protecting land movement paths and natural water systems at the same time is the best way to support nature and create peaceful coexistence between humans and wildlife.

Future Scope

In the future, this project can be expanded in many useful ways. More advanced sensors and better tracking technology can be added to monitor animal movement more accurately. Higher-quality water monitoring systems can help measure water levels, water purity, and seasonal changes with more detail.

This system can also be connected to real-time alert messages so that when animals move close to villages or farms, local people and forest officials receive warnings on mobile devices. Machine learning models can be improved to predict animal movement and water needs ahead of time, allowing proactive planning instead of reacting after problems occur.

The project can be scaled to larger areas, covering multiple forests and connecting corridors between different states or countries. Involving more communities, schools, and local groups can strengthen awareness and encourage people to participate in nature protection.

Finally, this system can support long-term conservation planning by helping governments create better policies for land use, road construction, forest protection, and water management. Over time, the data collected can build a valuable knowledge base to guide future environmental decisions.

In simple words, this study can grow into a strong real-world system that protects wildlife, supports water conservation, and helps humans live safely with nature

References

1. S. Singh et al., “IoT-based Smart Traffic Management System,” IEEE Access, 2023.

2. A. Kumar and R. Gupta, “Machine Learning for Intelligent Transportation Sys tems,” Springer, 2022.

3. P. Jain, “Real-Time Traffic Analysis using IoT,” International Journal of Computer Applications, 2021.